Predicting PVT and Thermodynamic Properties of Real Gases Using Peng–Robinson EOS and Machine Learning (`realgas-ml` v0.3.0)

Satwik Gupta (Roll No. `2025CH11012`) & **Udhav Jaipuria** (Roll No. `2025CH11729`)

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1. Introduction and Thermodynamic Equations

In this project for our Chemical Engineering Thermodynamics course, we built `realgas-ml` (v0.3.0) (`pyproject.toml`, `README.md`), a self-contained Python package and web app that predicts the pressure–volume–temperature (*PVT*), enthalpy, entropy, and fugacity properties of 15 pure real gases in the supercritical region. Instead of calling external property libraries like CoolProp or Cantera at runtime, our code generates *65,536* state points per gas (*983,040* points in total) directly from the Peng and Robinson (1976) cubic equation of state (PR76) combined with NASA 7-coefficient ideal-gas heat-capacity polynomials. For each gas, we train a four-output local radial-basis-function (`LocalRBFRegressor`) model over a normalized temperature and log-pressure grid so that properties can be predicted quickly and accurately without solving a cubic equation every time.

In our analytical solver `realgas_ml/peng_robinson.py`, pressure P (Pa) is related to absolute temperature T (K) and molar volume v ($m^3\ mol^{-1}$) by the Peng–Robinson (1976) equation of state:

$$P(T,v) = (RT)/(v - b) - (a(T))/(v^2 + 2bv - b^2), \qquad a(T) = a_c[1 + \kappa(1 - \sqrt{((T)/(T_c))})]^2, \qquad b = 0.07780\ (RT_c)/(P_c)$$

where $R = 8.31446261815324\ J\ mol^{-1}\ K^{-1}$ is the universal gas constant, $a_c = 0.45724\ R^2T_c^2/P_c$ is the attraction parameter at the critical point, T_c and P_c are the critical temperature and critical pressure, and $\kappa = 0.37464 + 1.54226\ \omega - 0.26992\ \omega^2$ depends on Pitzer's acentric factor ω. Differentiating $a(T)$ with respect to temperature gives the derivative needed for enthalpy and entropy departure calculations, $(da)/(dT) = -fraca_c\ \kappa\ g(T)T_c\sqrt{(T/T_c)}$, where $g(T) = 1 + \kappa(1 - \sqrt{(T/T_c)})$. Using the dimensionless parameters $A = (a(T)P)/(R^2T^2)$ and $B = (bP)/(RT)$, Equation (1) is written as the standard cubic polynomial in terms of the compressibility factor $Z = (Pv)/(RT)$:

$$Z^3 + (B - 1)Z^2 + (A - 3B^2 - 2B)Z + (-AB + B^2 + B^3) = 0$$

The function `cubic_z(A, B)` in `peng_robinson.py` solves Equation (2) at each (T,P) point using `numpy.roots`, keeps only physically valid real roots with $|Im(r)| < 10^{-9}$ and $Re(r) > B$ (which ensures $v > b$), and picks the largest real root corresponding to the single-phase gas root.
To calculate real-gas enthalpy, entropy, and fugacity departures (Maginn, CBE 20260), we define the logarithmic term L:

$$L = ln[fracZ + (1 + \sqrt{(2)})BZ + (1 - \sqrt{(2)})B] = log1p[frac2\sqrt{(2)}\ BZ + (1 - \sqrt{(2)})B]$$

Computing L with `np.log1p` in `peng_robinson.py` avoids subtracting two nearly equal numbers at low pressures ($P \to 1\ bar$, where $B \ll 1$ and $Z \to 1$), which prevents floating-point rounding errors. Using L, the molar enthalpy departure h_dep, molar entropy departure s_dep, log fugacity coefficient $ln\varphi$, and fugacity f follow standard textbook departure formulas:

$$h_dep(T,P) = RT(Z - 1) + fracT(da)/(dT) - a(T)2\sqrt{(2)}\ b\ L, \qquad s_dep(T,P) = Rln(Z - B) + frac(da)/(dT)2\sqrt{(2)}\ b\ L$$

$$ln\varphi(T,P) = Z - 1 - ln(Z - B) - fracA2\sqrt{(2)}\ B\ L, \qquad f(T,P) = \varphi(T,P)\ P$$

The ideal-gas changes $h_ig(T)$ and $s_ig(T)$ are calculated in `ideal_integrals(T, info)` by analytically integrating the first five coefficients $(a_0,\ldots,a_4)$ of the NASA 7-coefficient polynomial $C_p°(T)/R = \sum_{k=0}^{4} a_k T^k$ from the reference state $T_ref = 300.0\ K$ and $P_ref = 10^5\ Pa$ ($1\ bar$), where we set $h_ig(T_ref) = 0$ and $s_ig(T_ref, P_ref) = 0$:

$$\tilde{h}(T) = R(a_0\ T + (a_1)/(2)T^2 + (a_2)/(3)T^3 + (a_3)/(4)T^4 + (a_4)/(5)T^5), \quad \tilde{s}(T) = R(a_0\ ln\ T + a_1\ T + (a_2)/(2)T^2 + (a_3)/(3)T^3 + (a_4)/(4)T^4)$$

We leave out the standard formation constants (a_5, a_6) and integrate across the low- and high-temperature NASA polynomial intervals ($[200, 1000]\ K$ and $[1000, 6000]\ K$, or $[300, 1000, 5000]\ K$ for SO_2) using `np.clip` so there is no jump at $1000\ K$ (which matters for H_2O since its temperature range goes up to $1100.06\ K$). Adding the departure terms gives the total molar enthalpy $h(T,P) = h_ig(T) + h_dep(T,P)$ and total molar entropy $s(T,P) = s_ig(T) - Rln(P/P_ref) + s_dep(T,P)$. Because the real-gas departure $h_dep(300\ K, 1\ bar)$ is not zero, total h and s are slightly non-zero at $300\ K$ and $1\ bar$.

2. Selected Gases and Operating Ranges

Table 1: Critical constants, operating ranges (P ∈ [1.0, 50.0] bar for all gases), reduced temperature bounds, and data sources for all 15 pure gases in `metadata.json`.

Fluid Name	Species	M ($g\ mol^{-1}$)	T_c (K)	P_c (bar)	ω (—)	T_{min} (K)	T_{max} (K)	$T_r = T/T_c$ Range	Critical / NASA Source
Helium	He	4.0026	5.1953	2.2832	*-0.38354*	300.00	600.00	*57.74–115.49*	CoolProp He / L10/90
Hydrogen	H_2	2.0159	32.9800	12.9300	*-0.21700*	300.00	600.00	*9.10–18.19*	Cantera (Poling) / TPIS78
Neon	Ne	20.1790	44.4000	26.6163	*-0.03549*	300.00	600.00	*6.76–13.51*	CoolProp Ne / L10/90
Nitrogen	N_2	28.0135	126.2000	33.9000	*0.03800*	300.00	600.00	*2.38–4.75*	Cantera (NIST) / TPIS78
CarbonMonoxide	CO	28.0101	133.4000	35.0000	*0.06600*	300.00	499.00	*2.25–3.74*	Cantera (NIST) / TPIS79
Argon	Ar	39.9480	150.6870	48.6300	*-0.00219*	300.00	600.00	*1.99–3.98*	CoolProp Ar / L 6/88
Oxygen	O_2	31.9988	154.5800	50.4300	*0.02200*	300.00	600.00	*1.94–3.88*	Cantera (NIST) / TPIS89
Methane	CH_4	16.0428	190.7000	46.3000	*0.01100*	300.00	600.00	*1.57–3.15*	Cantera (NIST) / L 8/88
CarbonDioxide	CO_2	44.0098	304.2000	73.9000	*0.22800*	334.62	600.00	*1.10–1.97*	Cantera (Yaws) / L 7/88
Ethane	C_2H_6	30.0690	305.4000	48.8000	*0.09900*	335.94	600.00	*1.10–1.96*	Cantera (NIST) / L 8/88
Propane	C_3H_8	44.0956	369.7000	42.5000	*0.15300*	406.88	628.81	*1.10–1.70*	Cantera (NIST) / L 6/90
Ammonia	NH_3	17.0305	405.5600	113.6339	*0.25569*	446.12	689.45	*1.10–1.70*	CoolProp NH_3 / TPIS89
n-Butane	$n\text{-}C_4H_{10}$	58.1222	425.6000	37.6000	*0.19900*	468.16	574.00	*1.10–1.35*	Cantera (NIST) / L 6/90
SulfurDioxide	SO_2	64.0638	430.6400	78.8658	*0.25613*	473.70	524.00	*1.10–1.22*	CoolProp SO_2 / J 6/61
Water	H_2O	18.0153	647.0960	220.6400	*0.34429*	711.81	1100.06	*1.10–1.70*	CoolProp H_2O / L 8/89

The fluid database in `realgas_ml/data/metadata.json` includes 15 common pure gases: noble gases (He, Ne, Ar), diatomic gases (H_2, N_2, O_2),

The fluid database in `realgas_ml/data/metadata.json` includes 15 common pure gases: noble gases (*He*, *Ne*, *Ar*), diatomic gases (H_2, N_2, O_2, *CO*), triatomic and polar gases (CO_2, SO_2, H_2O, NH_3), and straight-chain alkanes from methane to *n*-butane (CH_4, C_2H_6, C_3H_8, $n\text{-}C_4H_{10}$). The NASA7 heat-capacity coefficients are taken from Cantera 3.2.0 `nasa_gas.yaml` (`SHA-256: 4de6199d...8667e4db`, based on NASA TM-4513) under the BSD 3-Clause license included in `THIRD_PARTY_NOTICES.txt`. Critical constants (T_c, P_c, ω, M) are taken from Cantera 3.2.0 `critical-properties.yaml` (citing Lemmon et al., 2022, NIST WebBook SRD 69; Yaws, 2001; and Poling et al.) for nine gases, and from CoolProp fluid reference tables for Helium, Argon, Neon, Ammonia, SulfurDioxide, and Water.

Table 1 summarizes the temperature and pressure bounds we chose for each gas. Every fluid covers the same pressure range of $P \in [10^5, 5 \times 10^6]$ *Pa* (*1–50 bar*), while the minimum temperature is set to $T_{min} = max(300.0\ K, 1.10\ T_c)$. Keeping $T \geq 1.10\ T_c$ ensures that every state point stays in the single-phase supercritical region well above the critical temperature, avoiding two-phase vapor–liquid splitting and the infinite compressibility slope $(\partial v/\partial P)_T \to -\infty$ at the critical point. Even so, for heavier hydrocarbons like *n*-Butane ($P_c = 37.60$ *bar*), Propane ($P_c = 42.50$ *bar*), and Ethane ($P_c = 48.80$ *bar*), a pressure of *50 bar* corresponds to reduced pressures $P_r = P_{max}/P_c$ of *1.33*, *1.18*, and *1.02* at $T_r = 1.10$. In this region near the critical point, intermolecular attraction causes strong real-gas deviations ($Z \approx 0.65–0.85$) and steeper changes in h_dep and $ln\varphi$. On the other hand, light gases like *He*, H_2, and *Ne* are at very high reduced temperatures ($T_r \in [6.76, 115.49]$), where molecular repulsion dominates ($Z > 1$) and the property surfaces are almost flat.

3. Machine Learning Model and Software Implementation

Our machine learning pipeline in `train.py`, `realgas_ml/regressor.py`, and `realgas_ml/model.py` is designed to handle two practical issues: pressure varies by a factor of 50 across the domain, and both compressibility Z and fugacity coefficient φ must always be positive. Instead of feeding raw (T, P) values into the model, `train.py` and `model.py` scale any state $(T, P) \in [T_{min}, T_{max}] \times [P_{min}, P_{max}]$ onto the unit square $x = (x_1, x_2) \in [0, 1]^2$ using linear temperature scaling and logarithmic pressure scaling:

$$x_1(T) = frac T - T_{min} T_{max} - T_{min} \in [0, 1], \qquad x_2(P) = fracln(P / P_{min}) ln(P_{max} / P_{min}) \in [0, 1]$$

Scaling pressure logarithmically helps in two ways. First, because the ideal-gas entropy contains the term $-Rln(P/P_ref)$, using $ln\ P$ as the input coordinate turns that logarithmic curve into a straight line in x_2, which the polynomial part of our model fits exactly. Second, spacing grid points evenly in $ln\ P$ puts more training points at lower pressures (*1–10 bar*) instead of crowding them all near *50 bar*. The model is trained on the four-output target vector $y = [ln\ Z, h, s, ln\varphi]^\wedge \to p \in \mathbb{R}^4$. During prediction in `GasModel.predict`, we take the exponential of the predicted $ln\ Z$ and $ln\varphi$ values ($Z = exp(\hat{y}_1)$ and $\varphi = exp(\hat{y}_4)$) so that $Z > 0$ and $\varphi > 0$ are guaranteed, and then compute all other molar and total properties for n moles:

$$v = (ZRT)/(P), \quad V = nv, \quad \rho = (M)/(v), \quad H = n\hat{y}_2, \quad S = n\hat{y}_3, \quad f = \varphi\ P$$

As checked in `tests/test_properties.py` and `tests/test_app.py`, fugacity f is an intensive property and does not change when the number of moles n changes, whereas total volume V, total enthalpy H, and total entropy S scale linearly with n.
In `realgas_ml/regressor.py`, `LocalRBFRegressor` wraps SciPy's `RBFInterpolator` inside a standard scikit-learn estimator class. First, the four target columns are standardized to zero mean and unit variance with `StandardScaler`. For any query point $x \in [0, 1]^2$, the interpolator finds the $k = 48$ closest training points $x_{i1}, \ldots, x_{ik}$ and builds a local quintic radial-basis-function interpolation with a degree-2 quadratic polynomial trend and zero smoothing ($\lambda = 0$):

$$tildey(x) = \sum_{j=1}^{48} w_j\ \varphi(\|x - x_{ij}\|_2) + \sum_{\alpha 1 + \alpha 2 \leq 2} c_{\alpha 1,\alpha 2}\ x_1{}^{\alpha}{}_1\ x_2{}^{\alpha}{}_2, \qquad \varphi(r) = -r^5$$

where the weights $w_j \in \mathbb{R}^4$ and polynomial coefficients $c_{\alpha 1,\alpha 2} \in \mathbb{R}^4$ pass through all *48* local training points ($tildey(x_{ij}) = y_std(x_{ij})$) along with *6* polynomial side conditions $\sum_{j=1}^{48} w_j\ x_{ij,1}{}^{\alpha}{}_1\ x_{ij,2}{}^{\alpha}{}_2 = 0$ for $\alpha_1 + \alpha_2 \leq 2$. Trying to fit a global RBF model across all $N = 65{,}536$ points at once would require solving a *65,536 × 65,536* dense matrix (*sim 34 GB* of memory), which is not practical on a normal computer. Using only the $k=48$ nearest neighbors reduces each prediction to a fast k-d tree lookup and a small *54 × 54* linear system. Each saved model file (`realgas_ml/data/.joblib`, compressed with level 3) takes only *2.63–2.66 MB* on disk.
When temperature T and molar volume v are given instead of pressure, `GasModel.pressure(T, v)` in `model.py` solves $\hat{v}(T, P) - v = 0$ for P using Brent's root-finding method (`scipy.optimize.brentq`) over $[10^5, 5 \times 10^6]$ *Pa*. Because molar volume always decreases with pressure at constant temperature ($(\partial v/\partial P)_T < 0$) in the single-phase gas region, Brent's method reliably finds the pressure to within a relative error of 10^{-8} (tested in `test_properties.py`). We provided four ways to use the package: the Python API (`example.py`), the command-line tool `realgas-ml` (`realgas_ml/cli.py`), a script to export NPZ data to CSV (`export_data.py`), and an interactive Streamlit web app (`app.py`, configured with `.streamlit/config.toml` and `run_app.bat`) that supports unit conversions (*K*/°*C*, *bar*/*kPa*/*MPa*/*Pa*), *(T,P)* and *(T,v)* modes, pressure plots, and CSV downloads.

4. Model Testing, Accuracy, and Thermodynamic Checks

Table 2: Interior test (N = 2,048) and boundary test (N = 256) errors comparing our ML models against direct Peng–Robinson calculations for all 15 gases trained on the 256 × 256 (65,536-point) grid.

Fluid	Int v MAPE	Int v Max	Int f MAPE	Int f Max	Int h MAE	Int s MAE	Bnd v MAPE	Bnd v Max	Bnd f MAPE	Bnd f Max	Bnd h Max	Bnd s Max
Helium	*3.46e-10*	*4.94e-08*	*2.80e-10*	*3.90e-08*	*1.02e-08*	*2.55e-10*	*1.23e-09*	*1.38e-07*	*9.38e-10*	*9.68e-08*	*2.19e-06*	*1.22e-08*
Neon	*5.49e-10*	*7.84e-08*	*5.66e-10*	*8.28e-08*	*1.98e-08*	*3.14e-10*	*5.25e-10*	*2.59e-08*	*5.46e-10*	*2.56e-08*	*7.37e-07*	*1.73e-08*
Hydrogen	*6.47e-10*	*7.74e-08*	*6.10e-10*	*7.26e-08*	*3.23e-08*	*3.32e-10*	*8.94e-10*	*6.60e-08*	*7.94e-10*	*4.95e-08*	*2.44e-06*	*2.82e-09*
Water	*9.96e-10*	*8.60e-08*	*9.53e-10*	*7.32e-08*	*4.53e-07*	*4.37e-10*	*1.47e-09*	*3.34e-08*	*1.14e-09*	*2.36e-08*	*6.54e-06*	*6.53e-09*
Nitrogen	*1.16e-09*	*3.12e-07*	*1.06e-09*	*2.46e-07*	*4.16e-08*	*4.22e-10*	*1.84e-09*	*5.90e-08*	*1.83e-09*	*5.43e-08*	*2.09e-06*	*2.23e-08*
Oxygen	*1.30e-09*	*1.90e-07*	*1.10e-09*	*1.20e-07*	*6.17e-08*	*4.57e-10*	*2.20e-09*	*9.14e-08*	*2.09e-09*	*7.17e-08*	*1.77e-06*	*1.78e-08*
CarbonMonoxide	*1.37e-09*	*2.81e-07*	*1.11e-09*	*1.72e-07*	*4.30e-08*	*1.78e-10*	*2.07e-09*	*1.98e-07*	*1.54e-09*	*1.05e-07*	*1.95e-05*	*6.15e-08*
Argon	*1.39e-09*	*2.72e-07*	*1.19e-09*	*1.78e-07*	*6.19e-08*	*3.25e-10*	*1.87e-09*	*3.31e-08*	*1.81e-09*	*3.18e-08*	*8.65e-07*	*5.36e-09*
Methane	*1.85e-09*	*1.60e-07*	*1.45e-09*	*8.28e-08*	*1.16e-07*	*5.04e-10*	*1.06e-08*	*1.37e-06*	*7.71e-09*	*8.10e-07*	*3.61e-05*	*2.00e-07*
Ammonia	*2.27e-09*	*2.14e-07*	*2.09e-09*	*1.87e-07*	*4.53e-07*	*6.38e-10*	*5.90e-09*	*5.11e-07*	*3.24e-09*	*2.50e-07*	*4.70e-05*	*9.10e-08*

Fluid	Int v MAPE	Int v Max	Int f MAPE	Int f Max	Int h MAE	Int s MAE	Bnd v MAPE	Bnd v Max	Bnd f MAPE	Bnd f Max	Bnd h Max	Bnd s Max
CarbonDioxide	*4.78e-09*	*9.96e-07*	*2.71e-09*	*4.80e-07*	*3.78e-07*	*7.16e-10*	*2.47e-08*	*2.84e-06*	*9.46e-09*	*7.60e-07*	*8.98e-05*	*3.59e-07*
SulfurDioxide	*1.07e-08*	*1.98e-06*	*5.74e-09*	*7.47e-07*	*1.43e-06*	*2.40e-09*	*1.09e-08*	*4.29e-07*	*5.34e-09*	*1.62e-07*	*4.74e-05*	*8.64e-08*
Ethane	*1.73e-08*	*7.53e-06*	*5.38e-09*	*1.06e-06*	*8.64e-07*	*2.17e-09*	*5.28e-08*	*2.25e-06*	*1.39e-08*	*4.26e-07*	*1.75e-04*	*4.74e-07*
Propane	*1.85e-08*	*5.63e-06*	*4.65e-09*	*4.77e-07*	*1.09e-06*	*2.23e-09*	*4.66e-08*	*8.53e-07*	*1.09e-08*	*1.53e-07*	*6.82e-05*	*1.62e-07*
n-Butane	*4.08e-08*	*9.48e-06*	*6.99e-09*	*3.41e-07*	*2.16e-06*	*3.89e-09*	*8.28e-08*	*1.04e-05*	*1.12e-08*	*1.50e-06*	*4.10e-04*	*8.24e-07*
Global Worst Case	*4.08e-08*	*9.48e-06*	*6.99e-09*	*1.06e-06*	*2.16e-06*	*3.89e-09*	*8.28e-08*	*1.04e-05*	*1.39e-08*	*1.50e-06*	*4.10e-04*	*8.24e-07*

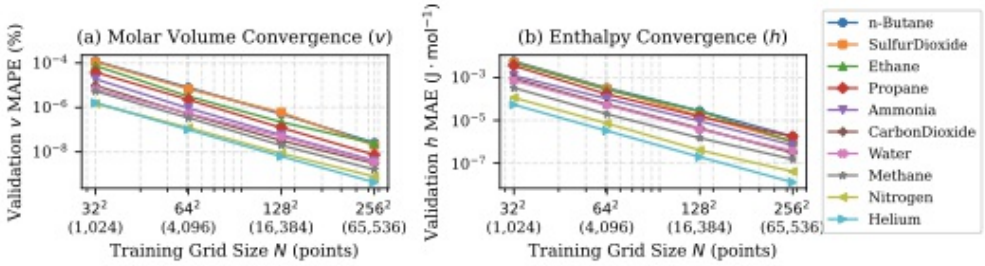


Figure 1: Validation learning curves (N_{val} = 512) for grid sizes N ∈ {32^2, 64^2, 128^2, 256^2} showing (a) molar volume MAPE and (b) molar enthalpy MAE.

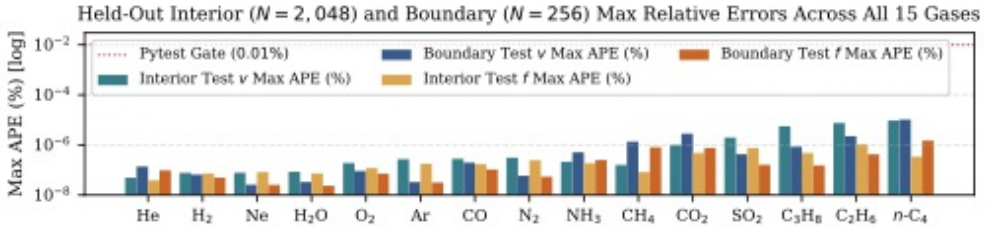


Figure 2: Comparison of interior test (N = 2,048) and boundary test (N = 256) maximum percentage errors on molar volume v and fugacity f across all 15 gases at N = 65,536.

Table 3: Validation learning-curve results (N_{val} = 512) across the four training grid sizes N ∈ {1,024, 4,096, 16,384, 65,536} for all 15 gases.

Fluid	v (1,024)	v (4,096)	v (16,384)	v (65,536)	p_{eff}(v)	h (1,024)	h (4,096)	h (16,384)	h (65,536)	p_{eff}(h)
Helium	*1.62e-06*	*1.01e-07*	*6.28e-09*	*4.33e-10*	*3.95*	*5.38e-05*	*3.28e-06*	*1.96e-07*	*1.29e-08*	*4.01*
Water	*1.50e-06*	*1.23e-07*	*8.55e-09*	*7.80e-10*	*3.64*	*7.28e-04*	*5.10e-05*	*3.83e-06*	*3.77e-07*	*3.64*
Neon	*1.82e-06*	*1.07e-07*	*8.70e-09*	*5.99e-10*	*3.86*	*7.12e-05*	*4.35e-06*	*3.56e-07*	*2.51e-08*	*3.82*
Hydrogen	*2.79e-06*	*1.83e-07*	*1.35e-08*	*1.28e-09*	*3.70*	*1.50e-04*	*9.14e-06*	*6.08e-07*	*4.87e-08*	*3.86*
Oxygen	*3.62e-06*	*2.60e-07*	*2.34e-08*	*1.88e-09*	*3.64*	*1.17e-04*	*6.68e-06*	*5.19e-07*	*5.63e-08*	*3.68*
Argon	*4.85e-06*	*2.65e-07*	*1.75e-08*	*1.73e-09*	*3.82*	*1.18e-04*	*9.52e-06*	*6.12e-07*	*6.33e-08*	*3.62*
CarbonMonoxide	*5.41e-06*	*3.83e-07*	*2.47e-08*	*2.07e-09*	*3.78*	*1.13e-04*	*7.06e-06*	*5.34e-07*	*5.56e-08*	*3.66*
Nitrogen	*5.45e-06*	*3.44e-07*	*2.00e-08*	*1.61e-09*	*3.91*	*1.14e-04*	*7.28e-06*	*4.08e-07*	*4.06e-08*	*3.82*
Ammonia	*6.60e-06*	*5.38e-07*	*4.87e-08*	*3.65e-09*	*3.61*	*1.21e-03*	*1.03e-04*	*9.37e-06*	*6.98e-07*	*3.59*
Methane	*9.70e-06*	*5.04e-07*	*3.24e-08*	*3.07e-09*	*3.87*	*3.29e-04*	*1.97e-05*	*1.51e-06*	*1.53e-07*	*3.69*
CarbonDioxide	*1.90e-05*	*9.29e-07*	*5.76e-08*	*4.04e-09*	*4.06*	*9.23e-04*	*5.69e-05*	*4.27e-06*	*3.08e-07*	*3.85*
SulfurDioxide	*3.85e-05*	*2.26e-06*	*1.22e-07*	*7.47e-09*	*4.11*	*5.14e-03*	*2.99e-04*	*1.64e-05*	*1.00e-06*	*4.11*
Propane	*7.52e-05*	*3.16e-06*	*2.20e-07*	*2.42e-08*	*3.86*	*3.54e-03*	*1.67e-04*	*1.49e-05*	*1.74e-06*	*3.66*
Ethane	*1.13e-04*	*6.76e-06*	*5.94e-07*	*2.00e-08*	*4.16*	*4.88e-03*	*2.75e-04*	*2.63e-05*	*1.02e-06*	*4.08*
n-Butane	*1.21e-04*	*7.91e-06*	*5.06e-07*	*2.60e-08*	*4.06*	*5.76e-03*	*3.27e-04*	*2.77e-05*	*1.79e-06*	*3.88*

In `train.py`, we used a fixed random seed (`np.random.default_rng(20261016 + index)`) to create four separate datasets for each gas: (i) the *256 × 256 = 65,536* training grid on *[0, 1]*^{2}; (ii) *512* random interior validation points to track learning curves at grid sizes of $N \in 32^2, 64^2, 128^2, 256^2$; (iii) *2,048* random interior test points; and (iv) *256* boundary test points (*64* random points along each of the four edges $x_1 \in 0,1$ and $x_2 \in 0,1$). For positive properties (*v* and *f*), we report Mean Absolute Percentage Error (*MAPE*) and Maximum Absolute Percentage Error (*Max APE*), calculated from the log predictions as $100\ |\exp(\hat{y} - y) - 1|$. For *h* and *s* (which pass through zero near *300 K* and *1 bar*), we report Mean Absolute Error (*MAE*) and Maximum Absolute Error (*Max AE*).

Table 2 and Figure 2 show that across all 15 gases on both interior and boundary test sets, the largest percentage error in molar volume is 1.0365×10^{-5}% (*n*-Butane boundary), the largest percentage error in fugacity is 1.4985×10^{-6}% (*n*-Butane boundary), the largest error in molar enthalpy is 4.0967×10^{-4} *J mol*$^{-1}$, and the largest error in molar entropy is 8.2417×10^{-7} *J mol*$^{-1}$ *K*$^{-1}$. All of these worst-case errors are 1,000 to 10,000 times smaller than the pass/fail tolerances checked in `tests/test_properties.py` (*Max APE < 0.01%* for *v* and *f*, *Max AE < 0.5 J mol*$^{-1}$ for *h*, and *Max AE*

$< 0.005\ J\ mol^{-1}\ K^{-1}$ for s).

The differences in error between gases in Table 2 make physical sense. Because our local RBF model includes a quadratic polynomial ($m=2$), it fits linear and quadratic trends exactly, so the remaining interpolation error depends on how sharply the property surface curves (third- and higher-order derivatives). For light gases like Helium ($T_r \geq 57.7$), Neon ($T_r \geq 6.76$), and Hydrogen ($T_r \geq 9.10$), the compressibility factor $Z(T,P)$ is very close to 1 and almost flat, giving tiny interior volume MAPEs between $3.46 \times 10^{-10}\%$ and $6.47 \times 10^{-10}\%$. Water also has a very small interior volume MAPE ($9.96 \times 10^{-10}\%$) despite being a polar molecule because $50\ bar$ is much lower than its critical pressure ($P_{max}/P_c = 50 / 220.64 = 0.2266$ at $T \geq 711.81\ K$), so steam behaves fairly smoothly over this range. In contrast, n-Butane, Propane, and Ethane reach $P_{max}/P_c \in [1.02, 1.33]$ near $T_{min} = 1.10\ T_c$, where $Z(T,P)$ bends more steeply near the critical region. Even for those heavier hydrocarbons, boundary errors are only $2–5\times$ larger than interior errors.

Table 3 and Figure 1 show how the validation error drops as we increase the grid size from 32×32 ($1{,}024$ points) up to 256×256 ($65{,}536$ points). Every time the number of grid points per side is doubled (which cuts the grid spacing Δx in half), the validation $MAPE$ and MAE drop by a factor of $12.2\times$ to $17.9\times$, giving an effective convergence rate of $p_eff \in [3.59, 4.16]$. Going from $1{,}024$ points to $65{,}536$ points reduces the prediction errors by more than $2{,}000\times$ to $5{,}600\times$.

To make sure our Peng–Robinson equations are thermodynamically consistent, `tests/test_properties.py` runs 47 automated tests across all 15 gases at $T_mid = (T_{min} + T_{max})/2$ and $P = 2.0\ MPa$ using central finite differences ($\Delta T = 0.01\ K$, $\Delta P = 100.0\ Pa$). These tests verify five standard thermodynamic relations from class to within relative tolerances of 10^{-10} to 2×10^{-5}:

$$g_dep = h_dep - T\,s_dep = RT\ln\varphi,\ ((\partial h)/(\partial P))_T = v - T((\partial v)/(\partial T))_P,\ ((\partial s)/(\partial P))_T = -((\partial v)/(\partial T))_P,\ ((\partial \ln f)/(\partial P))_T = (v)/(RT),\ ((\partial h)/(\partial T))_P = T((\partial s)/(\partial T))_P$$

5. Conclusion and Limitations

Our `realgas-ml` v0.3.0 project shows that combining log-pressure scaling and log-target transforms ($\ln Z$, $\ln\varphi$) with local quintic RBF interpolation ($k=48$, $m=2$) reproduces analytical Peng–Robinson and NASA7 property calculations across 15 pure gases to 8–10 significant digits ($Max\ APE \leq 1.04 \times 10^{-5}\%$). By avoiding repeated cubic root solving during forward lookups and using Brent's method for inverse $P(T,v)$ calculations, the package gives fast, self-contained predictions for coursework and process calculations. Still, two important limitations noted in `README.md` and `model.py` should be kept in mind. First, because the four outputs $[\ln Z, h, s, \ln\varphi]$ are fitted as separate columns, the ML model's derivatives satisfy the Maxwell relations in Equation (10) up to the $sim\ 10^{-8}$ interpolation error rather than exactly; calculations that need exact derivative identities should call `peng_robinson.properties` directly. Second, our low test errors measure how closely the ML model matches the PR76 equation of state, not experimental lab data; PR76 itself is an approximate cubic equation of state and can have noticeable physical error for strongly polar fluids (H_2O, NH_3) or light quantum gases (He, H_2).

References and Project Links

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7. Project Repository (`realgas_ml`): `README.md`, `pyproject.toml`, `peng_robinson.py`, `regressor.py`, `model.py`, `train.py`, `app.py`, `cli.py`, `export_data.py`, `metadata.json`, `test_properties.py`, and `test_app.py`.